ORPHOGRAPH

How it works

A picture-first walk from Bitcoin down to your own receipt — each idea drawn, defined, and said four ways: for a child, a teen, an adult, and a senior.

Orphograph turns a file into a checkable timestamp anchored to Bitcoin. Your file never leaves your device — only its fingerprint travels.

STEP 1

What is Bitcoin?

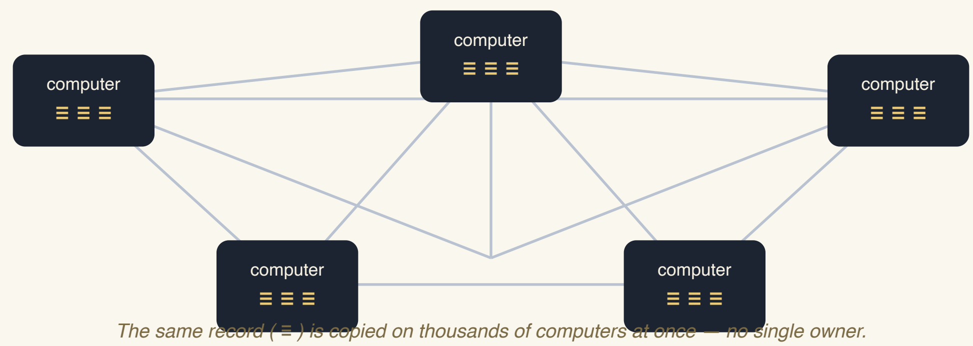


Figure 1 — Bitcoin is one shared record, held identically by computers all over the world.

Definition. Bitcoin is a public network of thousands of independent computers that all keep the exact same list of records. No person or company owns it, and because everyone holds a copy that the others constantly check, rewriting the past is practically impossible.

For a child Imagine the whole class keeps the same notebook at the same time. If one kid erases a page, everyone else's notebook still shows the truth. Bitcoin is a giant shared notebook.

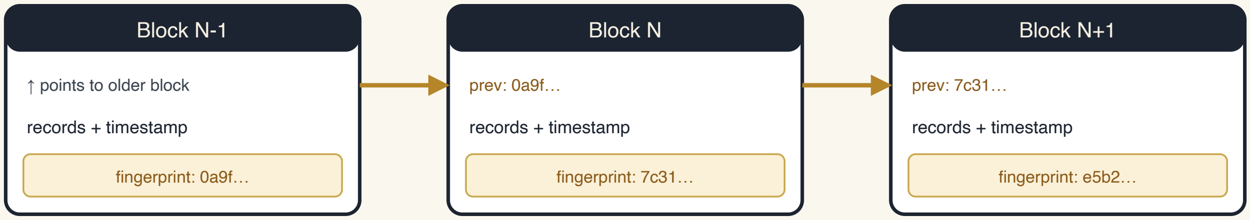
For a teen It's one record kept on thousands of computers at once. They keep checking each other, so you can't secretly rewrite something that already happened.

For an adult A decentralized ledger replicated across independent nodes worldwide; consensus among them makes recorded history extremely costly to alter.

For a senior Picture a public register kept in thousands of town halls simultaneously. To forge an entry you'd have to change every copy at the same moment — which no one can do.

STEP 2

What is a block?



Each block carries the fingerprint of the one before it — the "chain" in blockchain.

Figure 2 — Blocks are locked together; change an old one and every fingerprint after it stops matching.

> **Definition.** Records are bundled into **blocks**. Each block is time-stamped and carries the fingerprint of the block before it, linking them into a chain — the "blockchain." Alter an old block and its fingerprint changes, breaking every link after it, so tampering is instantly visible.

For a child Blocks are like train cars hooked together. Each car is locked to the one in front. Swap a car and the hooks don't match — everyone notices.

For a teen Every block points back to the one before it with its fingerprint. Edit an old block and all the following fingerprints stop matching — it shows immediately.

For an adult Blocks batch transactions with a timestamp plus the prior block's hash; that hash-linkage makes the ledger append-only and tamper-evident.

For a senior Each page of the ledger is sealed and references the seal of the page before. Alter an old page and every later seal no longer fits.

STEP 3

What is a hash?

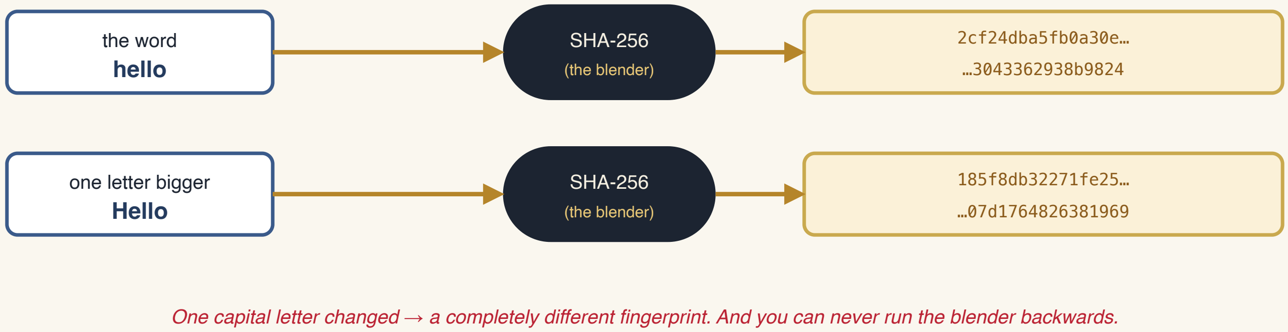


Figure 3 — A hash turns any input into a fixed 64-character fingerprint. Same in → same out; tiny change → total change; one-way only.

> **Definition.** A **hash function** (Bitcoin and Orphograph use SHA-256) takes any data — a word or a huge file — and produces a fixed 64-character fingerprint. The same input always yields the same fingerprint; change a single character and it changes completely; and you cannot work backward from the fingerprint to the data.

For a child A magic blender: put anything in, out comes a special code. The same thing always makes the same code — but you can never un-blend the code back into the thing.

For a teen It squashes any file into a short code. Same file = same code every time; change one letter and the code looks totally different. You can check a match, but never reverse it.

For an adult A one-way cryptographic digest: deterministic, fixed-length, collision-resistant, with an avalanche effect — flipping one input bit flips about half the output.

For a senior Like a unique wax seal for a document. The same document always makes the same seal; the slightest edit makes a different seal; and you can't recreate the document from the seal.

STEP 4

What is a Merkle tree?

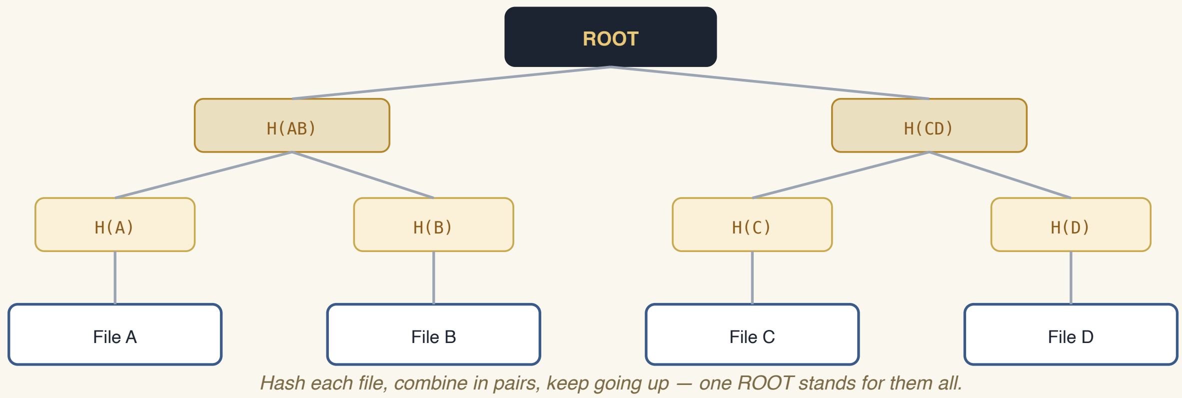


Figure 4 — A Merkle tree folds many files into one root fingerprint; change any file and the root changes.

Definition. To fingerprint many files at once, you hash each file (the **leaves**), then hash them together in pairs, and keep combining upward until a single fingerprint — the **root** — represents them all. Change any one file and the root changes. It also lets you prove one file belongs to the set using just a few fingerprints, without revealing the others.

For a child A family tree of codes. Each toy gets a code, the codes hold hands two by two, all the way up to one big code at the top that stands for every toy.

For a teen Hash each file, pair them up, hash again, repeat until one "root" code is left. That root sums up everything; touch any file and the root won't match.

For an adult A binary hash tree whose root commits to every leaf. Inclusion proofs are logarithmic — prove membership with a handful of sibling hashes, no need to reveal the whole set.

For a senior Like folding many signed pages into one master seal. That single seal stands for all the pages, and you can show one page belongs under it without opening the rest.

STEP 5

How your file becomes a receipt

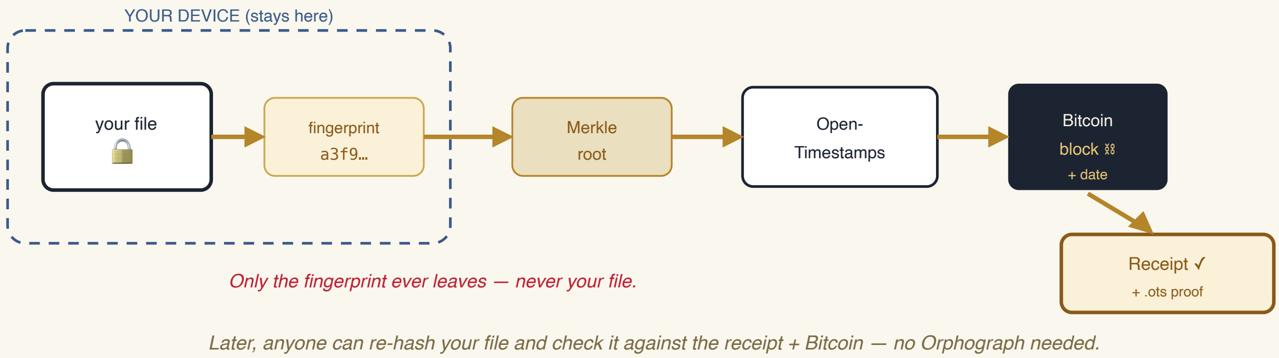


Figure 5 — File → local hash → Merkle root → OpenTimestamps → Bitcoin block → your receipt.

> **Definition.** Orphograph hashes your file **on your own device** — the file itself never leaves. That fingerprint (with others) forms a Merkle root, which is submitted to OpenTimestamps and committed into a Bitcoin block. Your receipt plus its **.ots** proof lets anyone later confirm this exact file existed, unchanged, by that block's time.

For a child The computer takes a secret code of your drawing — not the drawing — and pins it to the world's giant notebook with today's date. Later you can show your drawing is the very same one from that day.

For a teen Your file's fingerprint gets locked into Bitcoin's timeline. Anyone can check that fingerprint later to prove the file existed then and hasn't been changed.

For an adult Local hash → Merkle root → OpenTimestamps → Bitcoin. The .ots proof anchors existence-and-integrity to a block height; it attests time and immutability, not authorship or ownership.

For a senior We take the seal of your document and record that seal in a permanent public register, with the date. Anyone can later confirm your document matches that dated seal — proof it existed then and hasn't changed.

Where Orphograph fits among the ways people prove things

People already have ways to say "this existed, and it's mine, and it hasn't changed." Each asks you to **trust** something different, and each treats your **file** and your **privacy** differently. Here's the honest comparison — by category, not by brand.

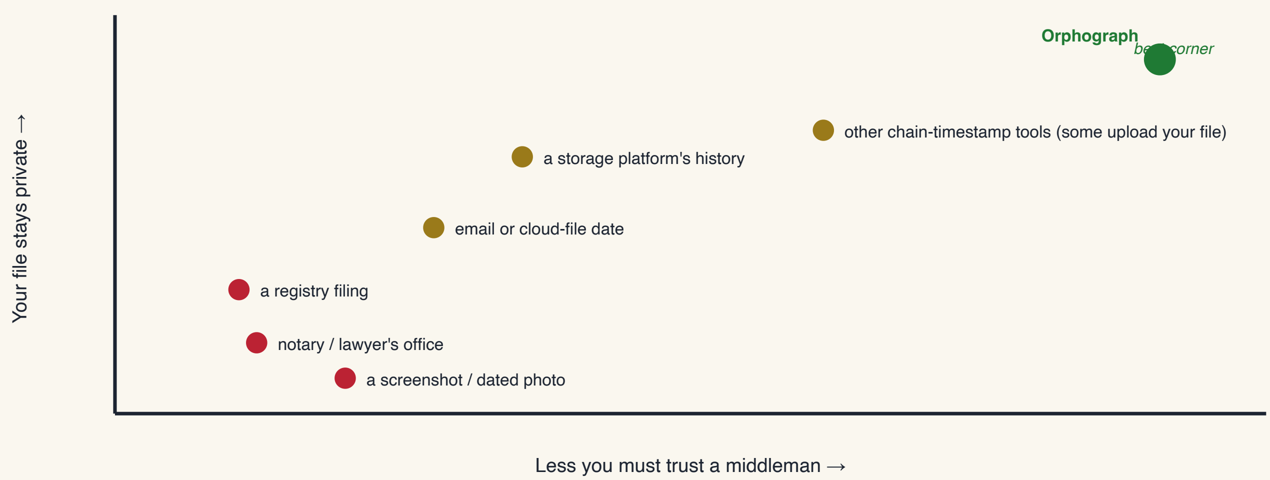


Figure 6 — Two things that matter: how little you must trust a middleman, and whether your file stays with you. Orphograph aims for the top-right.

Way of proving it	Who you must trust	Does your file leave you?	Can the record be changed or lost later?	Can anyone verify it, forever?
A notary / lawyer's office	a person & their office	often (they keep a copy)	records can be lost/destroyed	no — you go back to them
A registry filing	a government office	yes	slow, jurisdiction-bound	only through that office
A screenshot / dated photo	a device clock	no	trivially faked	no
Email or cloud-file date	the provider	yes	provider controls the clock	no
A storage platform's history	the platform	yes (they hold it)	they can edit / lose / close	only while they exist
Other chain-timestamp tools	varies (some upload files)	sometimes	depends on the chain	depends on the tool
Orphograph	math + Bitcoin (no single owner)	no — only a fingerprint	no — anchored & append-only	yes — even if Orphograph is gone

> **Definition.** Most ways of proving something existed rely on a **custodian** you must trust — a person, an office, or a company that could change the record, lose it, or shut down. Orphograph's difference is **minimized trust**: your file is hashed locally (privacy), the proof is anchored to Bitcoin (no single

> controller), and it can be checked by anyone, independently, even if Orphograph itself disappears. What it does **not** do is claim who made the file, who owns it, or that its contents are true.

For a child There are lots of ways to say "I made this first" — tell the teacher, take a photo, mail yourself a letter. Orphograph pins your secret code to a giant notebook the whole world keeps, so nobody — not even us — can move it later.

For a teen Other ways make you trust a person, an app, or a company that could change or lose your proof, or just shut down. Orphograph puts the proof on Bitcoin, which no one company controls — and only your fingerprint travels, never your file.

For an adult Where alternatives trust a custodian (notary, platform, provider clock), Orphograph minimizes trust: local hashing for privacy, OpenTimestamps anchoring to Bitcoin so there's no single point of control, and vendor-independent verification. It attests time + integrity, not authorship.

For a senior Instead of trusting one office or company to keep your proof safe, Orphograph records it in a permanent public register that no one owns and anyone can check — and it only ever sends a seal, never your document.

The same idea, per industry

One mechanism, many uses. Where a checkable "this existed, unchanged, by this date" is worth having:

Writers & authors	Pin a manuscript's fingerprint the day it's finished — evidence of who held it first.
Photographers	Timestamp a shot at capture, before it's reposted, scraped, or fed to a model.
Designers & artists	Prove a design existed before a look-alike appeared.
Musicians	Date a demo or a lyric sheet the moment it's written.
Lawyers & notaries	Seal a signed agreement or an evidence file to a fixed date — capture and timestamp only, never legal advice.
Plumbers & electricians	Timestamp before/after job photos to settle "it was never done like that" disputes.
Auto body shops	Lock damage photos to a date for cleaner, less-disputed insurance claims.
Researchers & AI teams	Anchor a dataset's provenance and license bundle so auditors can verify it later.
Software teams	Prove a release or build existed at a given moment.
Consultants & agencies	Stamp a deliverable at handoff so scope can't be quietly rewritten afterward.

What a receipt does and doesn't say. An Orphograph receipt proves that a file existed in exactly this form by a recorded date, and that it has not changed since. It does **not** prove who created the file, who owns it, or that its contents are true — and it is not, on its own, a legal determination. The Bitcoin-chain half of verification is carried by the receipt's **.ots** proof and can be checked with any OpenTimestamps client, independently of Orphograph.